

BEFORE

The ancestors of modern humans (*Homo sapiens*) colonized Africa, Europe, and western Asia.

LIFE IN THE FREEZER

The last glaciation (colder period) of the Ice Age, when many areas were covered in ice, lasted from 100,000 to 50,000 years ago. During this period, sea levels were far below modern levels and **Siberia and Alaska were linked by land**.

It was also **drier than today** and tropical climates were slightly cooler **<< 22–23**. This was followed by a slightly warmer period before a return to extreme cold about 18,000 years ago.



ICY LANDSCAPE

ORIGINS OF MODERN HUMANS

The original ancestors of modern humans evolved south of the Sahara Desert in tropical Africa. The scattered **human population was very small**, and groups developed in isolation from each other.

EARLY MAN ON THE MOVE

Early *Homo erectus* fossils **<< 19** indicate that they had settled in western Europe by 800,000 years ago. Neanderthals **<< 19** spread into Europe and western Asia by 200,000 years ago.

FLINT HAND-AXES

This **technology**, developed in Africa 2.5 million years ago, was used for millions of years **<< 17**.



FLINT HAND-AX

100,000 The number of years since small groups of humans began to leave Africa. By 60,000 years ago genetically modern *Homo sapiens* were colonizing the Earth.



Mammoth cave painting

The walls of Lascaux Cave in France are alive with bison, mammoth, wild oxen, and stag. Cro-Magnon artists (see p.26–27) painted these powerful, ageless images in this Ice Age treasure trove of art some 17,000 years ago.

Out of Africa

Every human today is the descendant of a small group of modern humans who left Africa around 60,000 years ago to explore the planet. We can see the legacy of these journeys today in the diversity of races and cultures around the world.

Sixty thousand years ago, modern humans (*Homo sapiens*) were confined to tropical Africa and a small part of southwestern Asia. These were people with the same physical and mental abilities as ourselves, hunter-gatherers capable of adapting to any environment on Earth, be it one with nine-month winters and subzero temperatures, or steamy tropical rainforests. Then, during the last cold period of the last Ice Age, the most significant of all human migrations out of Africa began. Toward the end of the Ice Age 15,000 years ago, this vast population movement was complete. Late Ice Age hunting bands had settled all of mainland Africa and Eurasia and had crossed, or were about to cross, into the Americas. *Homo sapiens* had mastered tropical waters with canoes or rafts, had drifted or paddled to New Guinea and Australia, and penetrated as far south as Tasmania.

Survival of the fittest

Earlier forms of humans such as *Homo habilis* and *Homo erectus* had long vanished from Earth, forced into extinction on marginal lands where food was not plentiful, or killed by the newcomers, with whom they could not compete.

Colonizing the planet was not a deliberate project, undertaken by men and women set on occupying new lands or exploring the world that lay beyond their hunting territories. Rather, the complex population movements that took modern humans to the limits of the harsh late Ice Age world came about as a result of the necessities of hunting and plant collecting in a great diversity of natural environments. In more northern climates, meat was the staple food, while tropical and temperate groups made considerable use of wild plant foods. The secrets to survival were adaptability—the ability to adjust to sudden changes in climatic conditions by technological innovation—and sheer ingenuity, mobility, and opportunism. People responded to food shortages, drought, or extreme cold by moving elsewhere in a world where the total global population was perhaps no more than five million people, scattered in small groups over hunting territories large and small.

Many people may not have encountered more than a few dozen fellow humans during their lives, although we can only speculate about this, as the population figures can only be educated guesses.

Evidence of migration

Dozens of archaeological sites—caves, rock shelters, open camps, and huge garbage heaps, or “middens,” of seashells and freshwater mollusks—document the great journeys made as humans spread around the globe. Klasies River Mouth in South Africa is one such site where caves were used as shelter by modern humans about 120,000 years ago, showing that by that date the first modern people had traveled from their origins in northeastern Africa (see pp.18–19).

The techniques of molecular biology are another way in which we can learn more about the movement of these early humans. By comparing certain strands of DNA (the substance found in every human cell that determines the characteristics we inherit), we can

1 MILLION The estimated human population of Earth 500,000 years ago.

work out how Earth was colonized by *Homo sapiens*, and when splits in the population occurred. This was a complex process involving constant movement by small numbers of people. We are only just beginning to comprehend the process of colonization, but one thing seems certain: all non-Africans are descended from what American biologist Stephen J. Gould once called “a single African twig” on the human family tree.

All people alive today have their ultimate roots in the so-called “African Eve” of some 150,000 years ago. This name stems from the fact that MtDNA (mitochondrial DNA) was passed from mother to offspring through every generation since the first *Homo sapiens*. We all share genetic information with “Eve,” with each other, and with our ancestors (see p.27).

Human migration

This map shows key sites for our early ancestors, as well as the routes that *Homo sapiens* is thought to have taken from Africa around the world.



Big game hunters The people associated with Clovis hunted big game. Their presence in America about 12,000 years ago coincides with the extinction of several large species including mammoths, mastodons (a mammothlike species), and giant sloths.

The earliest known settlement in South America dates from about 13,000 years ago. Finds from the site, at Monte Verde in Chile, include stone tools for chopping, scraping, and pounding.